

Total Questions: 35

LEVEL – II

Time: 1.5 hr.
Total Marks: 70

SECTION - 1

- This section contains **FIFTEEN (15)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers.

SECTION - 2

- This section contains **EIGHT (08)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each correct answer you will get **TWO** marks, if all the options are correct. If the correct answers chosen are less than the actual correct answers, partial marking (basis on weightage) will be given.
- There are **NO NEGATIVE** marks for wrong answers.

SECTION - 3

- This section contains **NINE (09)** questions of **PASSAGE TYPE OR COMPREHENSION TYPE**.
- Under each paragraph there will be three questions that has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers.

SECTION - 4

- This section contains **THREE (03)** questions. These questions are **MATCHING TYPE BASED QUESTIONS**.
- For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers

Section - 1

1. A cube rests on a rough horizontal surface. Equal forces of 10 N act on it: one east, one west, and one upward. What is true?
 - A) Net force is positive
 - B) Cube moves upward
 - C) Cube remains at rest but pressure decrease
 - D) Cube moves east
2. While most metals are solids at room temperature due to strong metallic bonding, Mercury is a liquid. Which of the following best explains this exception from a structural perspective?

- A) The presence of a partially filled d-subshell prevents the formation of a stable lattice structure.
- B) Mercury atoms form diatomic molecules similar to non-metals like Nitrogen.
- C) Mercury has a completely filled 6s orbital which is highly stabilized by relativistic effects, weakening interatomic attraction.
- D) Mercury has a very low atomic mass compared to other transition metals in its period.

3. A student was observing two different slides under a microscope. Slide A contained cells with a prominent central vacuole and a rigid outer boundary, while Slide B contained cells with multiple small vacuoles and lacked a rigid outer boundary. If the student adds a highly concentrated salt solution to both slides, what is the most likely observation?
 - A) Both cells will burst due to the sudden entry of water into the cytoplasm.
 - B) Slide A will show the cell membrane shrinking away from the cell wall, while Slide B will simply shrink in overall size.
 - C) Slide B will remain unaffected because it lacks a cell wall to exert pressure.
 - D) Slide A will swell up and become turgid, while Slide B will undergo lysis immediately.
4. A child pushes a heavy cupboard but fails to move it. Another child joins and both push together; now it moves. What does this situation indicate about friction?
 - A) Friction suddenly disappears
 - B) Static friction has a fixed value
 - C) Static friction increases with applied force up to a limit
 - D) Sliding friction is always greater
5. Aqua Regia is a mixture of concentrated and in a 3:1 ratio. Why can it dissolve 'noble' metals like Gold, which neither acid can dissolve alone?
 - A) oxidizes the gold into ions, and provides chloride ions to form a stable complex.
 - B) The mixture creates a higher temperature that melts the gold.
 - C) The mixture neutralizes the 'noble' properties of the gold.
 - D) The acids react to form pure Carbon, which acts as a catalyst.
6. In an experiment to study the action of yeast, 10g of yeast was added to a bowl of sugar solution at 35°C (Bowl X) and another 10g was added to a bowl of plain distilled water at 35°C (Bowl Y). After two hours, Bowl X smelled of alcohol and was frothy, while Bowl Y showed no change. What conclusion can be drawn regarding the role of microorganisms here?
 - A) Yeast acts as a catalyst and does not require a food source to produce carbon dioxide.
 - B) Microorganisms like yeast require carbohydrates to perform fermentation and release energy.
 - C) Temperature is the only factor that determines the rate of microbial reproduction.
 - D) Yeast is a bacterium that converts water into alcohol in the presence of heat.
7. A sound source produces waves of frequency 250 Hz in air. The wave travels a distance of 1700 m. After traveling half the distance, it enters a medium where speed becomes 5 times that in air. Find the total number of complete waves reaching the observer. v_{air} (330 m/s)

A) 100	B) 500
C) 772	D) 802
8. During the process of destructive distillation of coal, four main products are

obtained. If a student collects a sample that is a thick, black, viscous liquid with a pungent smell, which of the following statements is FALSE regarding this specific substance?

- A) It is a mixture of about 200 different substances.
- B) It can be used as a starting material for manufacturing synthetic dyes and explosives.
- C) It is the primary source for the production of naphthalene balls.
- D) It is a pure form of carbon used in the manufacture of steel.

9. A farmer in a water-scarce region wants to cultivate a crop that requires precise watering near the roots to minimize evaporation and runoff. However, his soil is sandy and does not retain moisture well. Which irrigation system should he prioritize, and what is a potential drawback he must manage?

- A) Sprinkler system; it mimics rain but may lead to fungal growth on leaves.
- B) Moat (pulley system); it is cost-effective but leads to significant water wastage in sandy soil.
- C) Drip system; it saves water but the nozzles can get clogged by impurities in the water.
- D) Chain pump; it is traditional and efficient for large-scale sandy plains.

10. A block of mass 5 kg is pulled by a force at 30° above horizontal. Coefficient of friction is 0.4. What is the minimum pulling force required to just move the block? ($g = 10 \text{ m/s}^2$)

- A) 18.6 N
- B) 25 N

- C) 30 N
- D) 40 N

11. In a vacuum chamber, a candle is ignited using an internal heating element, and then a small, controlled amount of pure oxygen is pumped in. What would be the most striking difference between this flame and a flame on Earth?

- A) The flame would be long and thin because of the high oxygen purity.
- B) The flame would be spherical because the absence of gravity prevents hot air from rising (convection).
- C) The flame would immediately extinguish because a vacuum cannot sustain a middle zone.
- D) The flame would turn green due to the lack of nitrogen in the chamber.

12. A particular forest area has been declared a Biosphere Reserve. Within this reserve, there is a National Park and a Sanctuary. A group of researchers wants to study the "Endemic Species" of this region. Which of the following scenarios best describes an endemic species they might find?

- A) A species of wild mango that is found in various tropical forests across the entire continent.
- B) A population of Giant Squirrels that exists only in this specific biosphere reserve and nowhere else in the world.
- C) Migratory birds that visit the sanctuary every winter from Siberia to escape the cold.
- D) Common livestock like cows and goats that have been introduced by local villagers for grazing.

13. During an educational trip to a poultry farm and a fish hatchery, a student observes that hens lay eggs with hard shells, while fish lay jelly-like eggs in water. Which of the following is a correct comparison of their reproductive strategies?

- A) Both show internal fertilization, but only the hen is oviparous.
- B) The hen shows internal fertilization and is oviparous; the fish shows external fertilization and is oviparous.
- C) The hen shows external fertilization and is viviparous; the fish shows internal fertilization and is oviparous.
- D) Both are viviparous, but the fish provides more parental care to its offspring.

14. In an IVF (In Vitro Fertilization) procedure, the fertilization occurs outside the body. However, the development of the baby still occurs inside the mother's uterus. Based on this, how would you classify the resulting birth?

- A) It is an example of external fertilization and external development.
- B) It is an example of asexual reproduction because it happens in a lab.
- C) It is an example of internal development following artificial fertilization, and the human remains viviparous.
- D) It is an example of oviparous birth because the egg was handled outside the body.

15. While studying cell organelles, Rahul learns that "Organelle X" is responsible for releasing energy from food, while "Organelle Y" is responsible for trapping solar energy. He also notices that both organelles have their own DNA and

ribosomes. Identify X and Y and their presence in human cheek cells.

- A) X is Mitochondria, Y is Chloroplast; both are present in human cheek cells.
- B) X is Chloroplast, Y is Mitochondria; only Y is present in human cheek cells.
- C) X is Mitochondria, Y is Chloroplast; only X is present in human cheek cells.
- D) X is Ribosome, Y is Nucleus; neither is present in human cheek cells.

Section - 2

16. A block of weight 240 N is placed on a surface of area 12 m². An upward force of 60 N is applied. Which are correct?

- A) Effective thrust on surface = 180 N
- B) Pressure = 15 Pa
- C) Pressure decreases due to upward force
- D) Block lifts off the surface

17. Consider four fuels (P, Q, R, and S) with the following characteristics. Which of the following conclusions are scientifically sound?

Fuel	Ignition Temp (°C)	Calorific Value (kJ/kg)	By-products
P	5	50,000	CO ₂ , H ₂ O
Q	45	45,000	CO ₂ , H ₂ O
R	350	80,000	CO, SO ₂
S	60	15,000	Ash, CO ₂

- A) Fuel P is the most efficient but is extremely dangerous to store at room temperature (25°C).

B) Fuel Q is a better domestic fuel than Fuel S because it has a higher calorific value and leaves no residue.

C) Fuel R is the most "ideal" fuel because it has the highest calorific value.

D) Fuel S has a high ignition temperature but is inefficient due to its low calorific value.

18. The "Red Data Book" serves as a global inventory of the conservation status of various species. If a species is moved from the 'Endangered' list to the 'Extinct' list, what does this imply?

A) The last known individual of that species has died.

B) The species can no longer be found in its natural habitat or in captivity.

C) The species has successfully migrated to a different planet or biosphere.

D) Exhaustive surveys in known and/or expected habitats have failed to record an individual.

19. Which of the following phenomena are explained correctly?

A) Diffuse (irregular) reflection occurs from rough surfaces and strictly violates the fundamental Laws of Reflection at the microscopic level.

B) When an opaque object is brought closer to a point source of light, the size of the umbra cast on a fixed screen behind it increases.

C) An extended source of light (like a tube light) casts a shadow consisting of both an umbra and a penumbra, whereas a perfect point source casts only an umbra.

D) The pupil of the human eye appears black because the light entering it is almost

entirely absorbed by the internal tissues of the eye (like the choroid) after multiple internal reflections, with very little reflecting back out.

20. Water is the most common fire extinguisher, but it cannot be used for all types of fires. In which of the following scenarios is water ineffective or dangerous, and why?

A) On a petrol fire, because petrol is lighter than water and will continue to burn on top of the water layer.

B) On an electrical fire, because tap water contains dissolved salts and can conduct electricity, leading to shocks.

C) On a magnesium fire, because water can react with very hot metals to release highly flammable hydrogen gas.

D) On a wood fire, because water increases the ignition temperature of the wood, making it harder to put out.

21. A farmer is shifting from traditional farming to "Sustainable Agriculture." Which of the following practices should he adopt to preserve soil quality while maintaining high yields?

A) Practicing crop rotation to ensure the soil is not depleted of specific nutrients.

B) Using a mixture of organic manure and measured amounts of chemical fertilizers.

C) Implementing the drip irrigation method to prevent water-logging and soil salinity.

D) Continuous monoculture (planting the same crop every season) to specialize in one product.

22. Antibiotics are powerful medicines, but they must be used with extreme caution. Why?

- A) Overuse can lead to the development of "Antibiotic Resistance" in bacteria.
- B) They can kill beneficial bacteria present in the digestive tract of the body.
- C) Taking them for viral infections like the common cold is ineffective.
- D) They increase the rate of viral replication if taken during a fever.

23. Biodiversity refers to the variety of life on Earth. Which of the following factors are responsible for the loss of biodiversity?

- A) Illegal poaching and hunting of wild animals.
- B) Introduction of invasive alien species that outcompete native species.
- C) Climate change and global warming.
- D) Expansion of agriculture and urban areas leading to habitat fragmentation.

Section - 3

COMPREHENSION-1

A neutral conducting sphere A is mounted on an insulating stand and connected to a gold-leaf electroscope. A positively charged rod R is brought near A without touching. While R is still near, A is earthed momentarily. Then earthing is removed, followed by removal of R. The electroscope leaves are observed.

24. Immediately after earthing (with rod still present), the electroscope leaves:

- A) Diverge positively
- B) Diverge negatively
- C) Collapse
- D) First diverge then collapse

25. After the complete process, the nature of charge on A is:

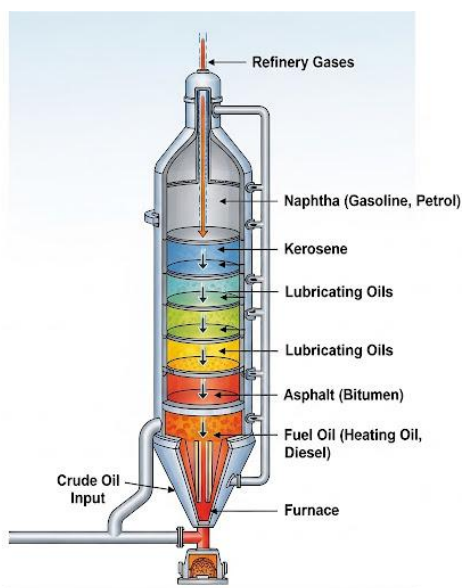
- A) Positive and uniformly distributed
- B) Negative and resides on outer surface
- C) Neutral but polarized
- D) Positive only near rod side

26. If instead the rod is removed before removing earthing, the final observation will be:

- A) Electroscope shows negative divergence
- B) Electroscope shows positive divergence
- C) No divergence (neutral)
- D) Continuous oscillation of leaves

COMPREHENSION-2

Crude oil, often called "Black Gold," is a complex mixture of various hydrocarbons ranging from small, volatile molecules to thick, heavy waxes. In its raw state, it has limited utility. To transform it into useful products like petrol, diesel, and aviation fuel, it must undergo a process called Fractional Distillation at a refinery. Inside a fractionating column, crude oil is heated to about 400°C. As the mixture vaporizes and rises through the tower, it cools. Different components (fractions) condense back into liquids at different heights based on their specific boiling points. Molecules with larger carbon chains have higher boiling points and settle at the bottom, while smaller molecules stay as gases or condense at the cooler top.



27. A technician at a refinery notices that the temperature at the top of the fractionating column has accidentally increased by 20°C . Which of the following is the most likely immediate consequence?

- A) The heavy residue (bitumen) will begin to vaporize and reach the top.
- B) The kerosene fraction will contain more heavy impurities, lowering its quality.
- C) More gasoline vapours will condense at the very bottom of the tower.
- D) The rate of crude oil intake will automatically double to compensate.

28. During a cold winter in a high-altitude region, a transport company finds that their vehicle fuel is thickening and clogging filters (a phenomenon called 'waxing'). Based on the refining process, which fuel change would logically solve this problem?

- A) Switching to a fuel fraction collected from a lower level of the tower.
- B) Switching to a fuel fraction with a higher carbon chain length.

C) Switching to a fuel fraction collected from a higher level of the tower (e.g., Kerosene-based).

D) Increasing the amount of Bitumen mixed into the diesel.

29. If a refinery needs to produce fuel specifically for jet engines (Aviation Turbine Fuel), which usually requires a boiling range of roughly 150°C to 250°C , which fractions are most likely involved in the blending process?

- A) Only P
- B) A blend of Q and R
- C) A blend of R and S
- D) Only S

COMPREHENSION-3

A group of students in a school laboratory performed an experiment. They took warm milk and added a small amount of curd to it, then kept it undisturbed for a few hours. After some time, the milk turned into curd. In another activity, one student left a piece of bread outside for a few days. Later, he observed greenish-black patches growing on it. The teacher explained that these patches were caused by microorganisms growing in moist conditions.

Meanwhile, a student fell sick after drinking contaminated water. The doctor said that harmful microorganisms had entered his body and caused the illness. From these observations, the teacher concluded that microorganisms can be both useful and harmful, depending on their type and conditions.

30. The greenish-black patches on bread are most likely:

- A) Algae growing in light
- B) Fungi growing in moist conditions
- C) Bacteria producing oxygen
- D) Viruses multiplying on food

31. Which condition mainly helped microorganisms grow on bread?

- A) Dry environment B) Moisture
- C) Strong wind D) Lack of air

32. Which of the following best describes the role of curd in the experiment?

- A) It acts as a source of heat
- B) It provides bacteria to convert milk into curd
- C) It prevents microbial growth
- D) It changes the colour of milk only

Section - 4

33.

COLUMN A	COLUMN B
A) Refraction of light	P) Light takes the path that requires the least time
B) Fermat's Principle	Q) The speed of light changes when entering a new material
C) Straight-line propagation	R) Light is traveling through a uniform, unchanging medium
D) Law of Reflection ($\angle i = \angle r$)	S) Light bounces back into the same medium without changing speed

- A) A-Q, b-P, c-R, d-S
- B) A-P, b-Q, c-S, d-R
- C) A-Q, b-R, c-P, d-S
- D) A-R, b-P, c-Q, d-S

34.

COLUMN -A	COLUMN-B
A) $\text{Mg(s)} + \text{HCl(dil.)}$	P) No visible reaction occurs at room temperature
B) $\text{Fe(s)} + \text{ZnSO}_4(\text{aq})$	Q) Rapid evolution of a gas that burns with a 'pop' sound
C) $\text{S(s)} + \text{H}_2\text{O(l)}$	R) Solution turns light green over time
D) $\text{Fe(s)} + \text{CuSO}_4(\text{aq})$	S) Non-metal remains inert to the liquid

- A) A-R, B-P, C-Q, D-S
- B) A-P, B-R, C-S, D-Q
- C) A-R, B-P, C-S, D-Q
- D) A-Q, B-P, C-S, D-R

35.

COLUMN -A	COLUMN-B
A) Nucleus	P) Produces energy
B) Mitochondria	Q) Controls cell activities
C) Cell membrane	R) Allows substances in and out
D) Ribosome	S) Protein synthesis

- A) A - P, B - Q, C - R, D - S
- B) A - Q, B - P, C - R, D - S
- C) A - R, B - S, C - Q, D - P
- D) A - Q, B - P, C - R, D - S